Bayu Febriansyah

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Education

University of Illinois Urbana-Champaign — Expected May 2027
M.S. in Aerospace Engineering; Current GPA: 3.83/4.00 — *Urbana, Illinois*

Bandung Institute of Technology (ITB) — Apr 2023
B.S. in Aerospace Engineering; Cum Laude; GPA: 3.64/4.00 — *Bandung, Indonesia*

Experience

3D Printing Engineer — May 2026 – Present
Illinois MakerLab — *Champaign, Illinois*
- Designed, optimized, and manufactured mechanical components, translating requirements into CAD models and functional parts; refined geometry, assembly, and printing parameters to improve manufacturability and reliability.

Undergraduate Researcher — Oct 2022 – Apr 2023
Lightweight Structures Laboratory — *Bandung, Indonesia*
- Developed and validated nonlinear finite element models in LS-DYNA; performed parametric and sensitivity analyses to characterize collapse mechanisms, energy absorption, and design tradeoffs in lightweight, thin-walled structures.

Flight Data Engineering Intern — Jul 2022 – Sep 2022
PT Aering — *Bandung, Indonesia*
- Designed and implemented a Python flight-reconstruction workflow, integrating coordinate correction, altitude and heading processing, and geospatial validation to generate X-Plane simulation input with positional accuracy within 8 m.

Mechanical Engineering Intern — Jun 2021 – Aug 2021
Gexcon — *Jakarta, Indonesia*
- Conducted ANSYS CFD simulations to characterize fire behavior and propagation under industrial plant accident scenarios, supporting hazard evaluation, plant risk assessment, and safety engineering consulting.

Selected Projects

Carbon-Epoxy Composite I-Beam | *Baur Research Group, UIUC* — 2026
- Led fabrication from the team's numerical design, adapting layup to prepreg, tooling, and equipment constraints; replaced foam-filled Nomex with aluminum honeycomb because uniform foam injection proved impractical.

Ultrathin Bistable Composite Boom | *Space Structures Laboratory, UIUC* — 2026
- Built MATLAB laminate stiffness and strain-energy models; screened 5,476 cases, identifying tradeoffs between transition energy and coiled geometry. Fabricated a boom and compared predicted stable shapes with DIC curvature.

Medical UAV System Architecture | *Systems Engineering Course, UIUC* — 2025
- Developed ConOps, requirements, functional decomposition and interface matrix for proposed medical UAV logistics: mission planning, airspace coordination, telemetry monitoring, fleet management and safety intervention.

Impact Analysis of Functionally Graded Crash Boxes | *Undergraduate Thesis, ITB* — 2023
- Compared five taper ratios under axial impact. Identified a graded-thickness multicell design with simulated specific energy absorption 124% above a conventional square tube and 11.8% above a conventional multicell tube.

Technical Skills

Programming and Data Processing: Python, C, MATLAB, Simulink, Mathematica, LaTeX, Microsoft Office.
CAD and Design: Siemens NX, SOLIDWORKS, Fusion 360, AutoCAD, PrusaSlicer, OrcaSlicer, Bambu Studio, Cura.
Engineering Simulation and Analysis: ANSYS, LS-DYNA, Abaqus, HyperMesh, Patran/Nastran, JavaFoil.
Languages: Indonesian (native) and English (fluent, TOEFL iBT 109/120, IELTS 8.0/9.0).

Honors, Awards, and Activities

Tau Beta Pi Engineering Honor Society (2026): Member; recognizes academic excellence and exemplary character.

Indonesia Endowment Fund for Education (LPDP) (2025): Awardee; Indonesian government graduate scholarship.

Dean's List, Bandung Institute of Technology (2023): Awardee; ranked among the top 5% of students consistently.

Memberships: Society for the Advancement of Material and Process Engineering (SAMPE), UIUC Chapter; American Institute of Aeronautics and Astronautics (AIAA), UIUC Chapter; Aerospace Graduate Student Advisory Committee (AeroGSAC); Indonesian Students Association (PERMIAS), UIUC; Indonesian Aerospace Engineering Alumni Association (IAP ITB).